

DEVELOPMENT OF A BONE IMPLANT

A Mechanically Optimized Solution for Affordable Orthopaedic Care



Project Team

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What is Limb Lengthening?

A medical procedure based on distraction osteogenesis where bone is surgically cut and gradually separated to stimulate new bone growth



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Current Solutions:

EXTERNAL FIXATION METHODS:

- traditional approach
- using an external frame attached to the bone with pins and wires.
- cost-effective
- widely used, especially in developing countries.
- disadvantages:
 - high risk of infection,
 - pain, scarring,
 - long treatment durations due to the prolonged use of external hardware.



Current Solutions:

INTERNAL LENGTHENING NAILS:

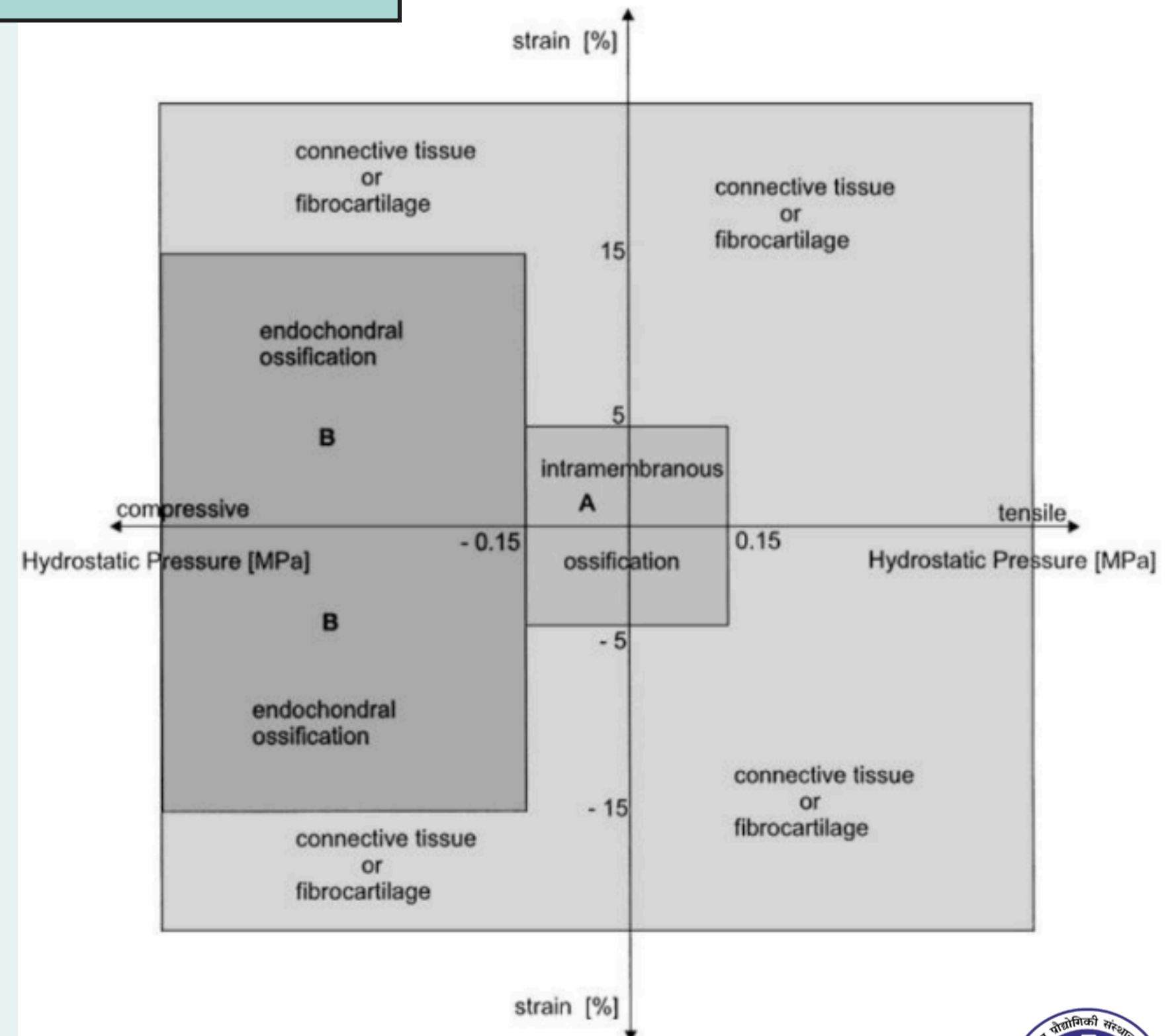
- Internal lengthening nails (implants) are modern, implantable devices placed inside the bone, offering a less invasive and more comfortable alternative.
- Examples include Fitbone, which uses a motorized system, and Precice Nail, which relies on a magnet-based mechanism.
- These implants have lower infection risks, cause less pain, and provide better patient comfort compared to external fixators.
- However, they are significantly more expensive and require advanced surgical expertise, making them less accessible in certain regions.



Background & Literature Review

STRESS-STRAIN CONDITIONS FOR BONE FORMATION:

- **Endochondral Ossification (B Region):**
 - Occurs under high compressive hydrostatic pressure (>0.15 MPa).
 - Happens when strain is moderate ($\sim 5-15\%$).
 - Seen in fracture healing with flexible fixation, where movement allows callus formation.
- **Intramembranous Ossification (A Region):**
 - Occurs at low strain ($<5\%$) and low hydrostatic pressure.
 - Seen in fracture healing with rigid fixation, where minimal movement allows direct bone formation.
- **Connective Tissue/Fibrocartilage Formation:**
 - High strain ($>15\%$) or excessive hydrostatic pressure prevents ossification.
 - This is undesirable as it leads to non-union or delayed healing.



Background & Literature Review

EXISTING SYSTEMS:

- **Fitbone (Orthofix):** Fully implanted with internal motor, remotely controlled.
- **Precice Nail (NuVasive):** Magnet-based system for non-invasive adjustment.



BIOLOGICAL PRINCIPLES:

- **Distraction osteogenesis** relies on controlled mechanical strain to stimulate bone regeneration.
- **Optimal conditions:** 0.5–1.5 mm/day distraction rate with proper latency period.



Problem Statement

PROJECT AIM:

Develop an affordable limb-lengthening implant using motor-driven actuator, planetary gear system, and lead screw mechanism

KEY CHALLENGES:

- High-end materials and complex electronics make existing systems prohibitively expensive.
- Need for controlled, progressive bone elongation without complications.
- Device must withstand physiological loads while maintaining precision.
- External control mechanism needed without invasive procedures.
- Balance between cost reduction and maintaining safety/functionality.

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SYSTEM ARCHITECTURE:

Rotational Motion: 6V DC motor with hall-effect encoder for precise rotational control.

Slow Down RPM: Three-stage planetary gear system (166.375:1 ratio) for torque amplification.

Rotation to Translation Motion: Lead screw (1.25mm pitch) for converting rotary to linear motion.

Controlling the Motion: Arduino-based closed-loop PID control system for positioning accuracy.

- **Manufacturing Approach:** Leveraging additive manufacturing and local production methods like 3D printing, LBM, etc.

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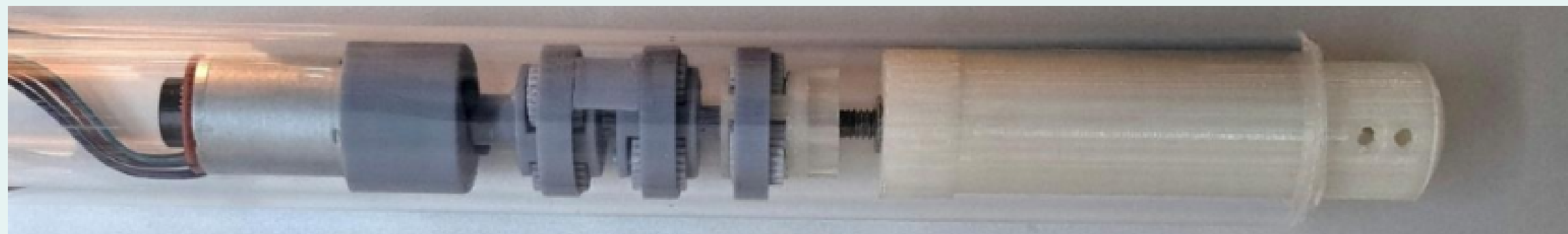
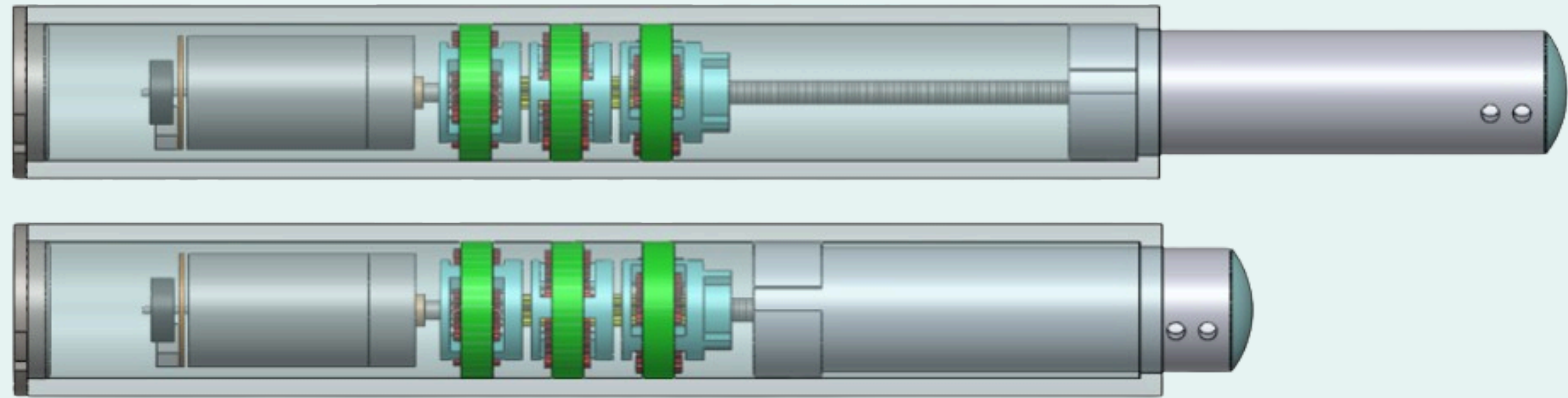
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Working Mechanism and Design

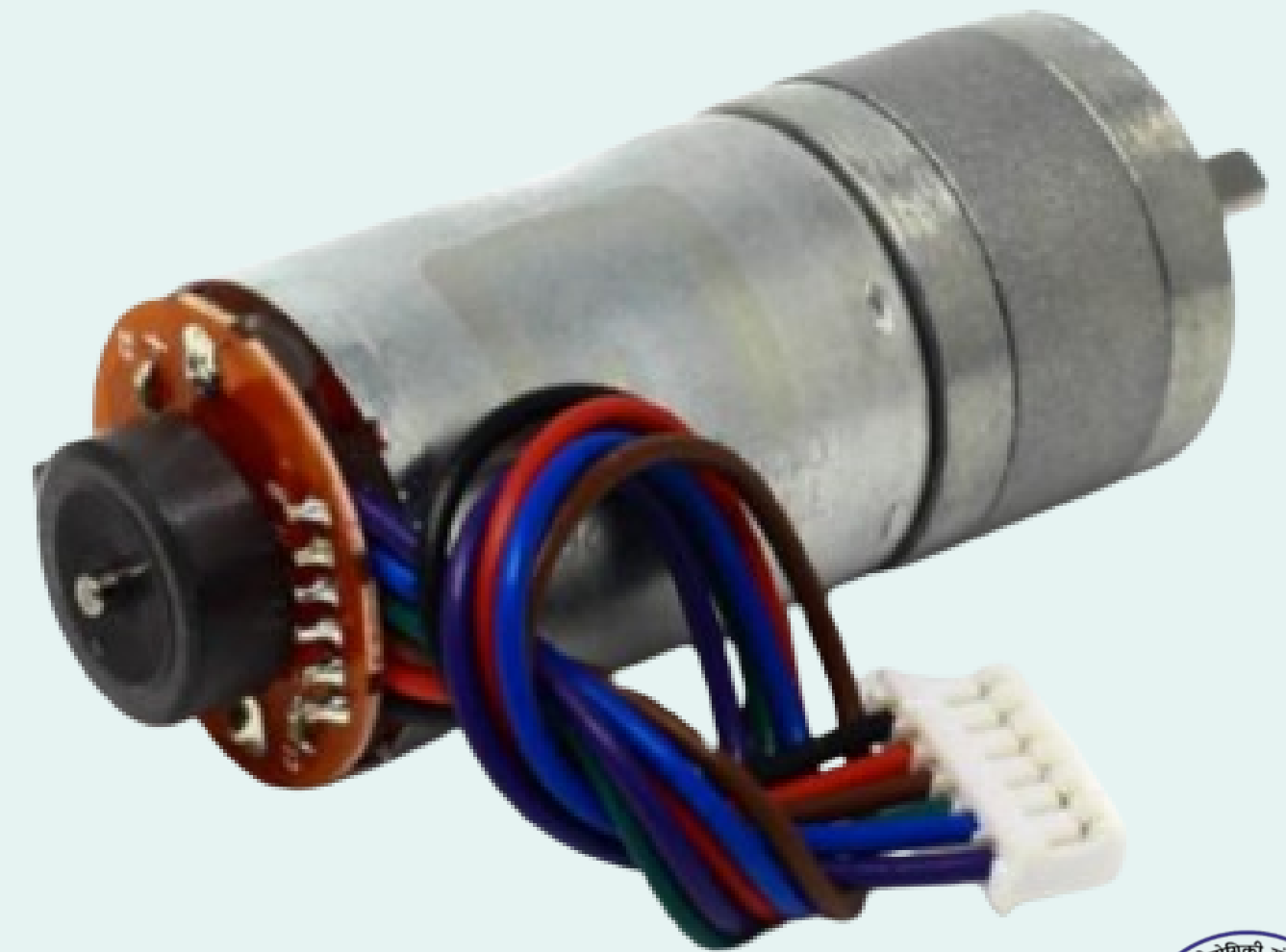


Working Mechanism and Design

The setup is powered by a DC motor and a three-stage planetary gear system. The motor provides high-speed rotary input, which is reduced in speed and amplified in torque through the three planetary stages. This enhanced torque is then transferred to a lead screw, converting the rotary motion into precise linear movement. As the lead screw rotates, it moves the attached extruder back and forth along the actuator's axis. This

1. DRIVING MOTOR WITH HALL-EFFECT ENCODER:

The driving motor used in this mechanism is a 6V DC motor rated at 260 RPM and equipped with a Hall-effect encoder. This motor provides the rotational input required to power the multi-stage planetary gear system. The built-in Hall-effect encoder plays a critical role in motion control by providing real-time feedback on the motor's speed and position.



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Working Mechanism and Design

2. PLANETARY GEAR SYSTEM:

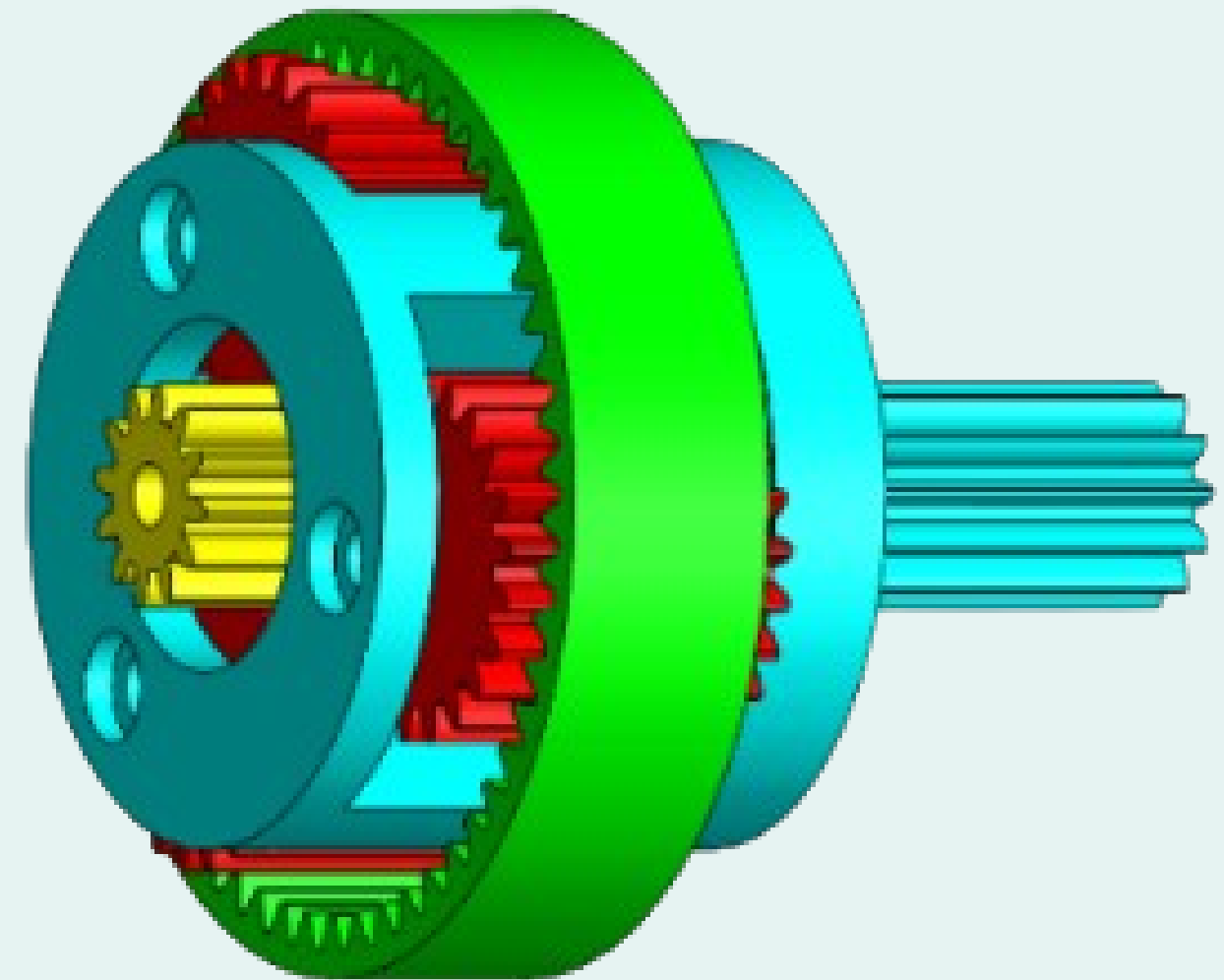
The planetary gear system is designed to reduce the rotational speed (RPM) while maintaining high torque efficiency.

This gear system operates using three main types of gears:

I. Sun Gear: Positioned at the center of the gear assembly, the sun gear receives input directly from the motor. It has 12 teeth with a module of 0.5 mm.

II. Ring Gear: This is the stationary outermost gear that encases the entire gear system. It has 54 teeth and serves as the structural boundary of the planetary setup.

III. Planet Gears: Located between the sun gear and the ring gear, these gears transmit power from the sun gear to the output shaft at a reduced speed. Each planetary gear has 18 teeth.



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Working Mechanism and Design

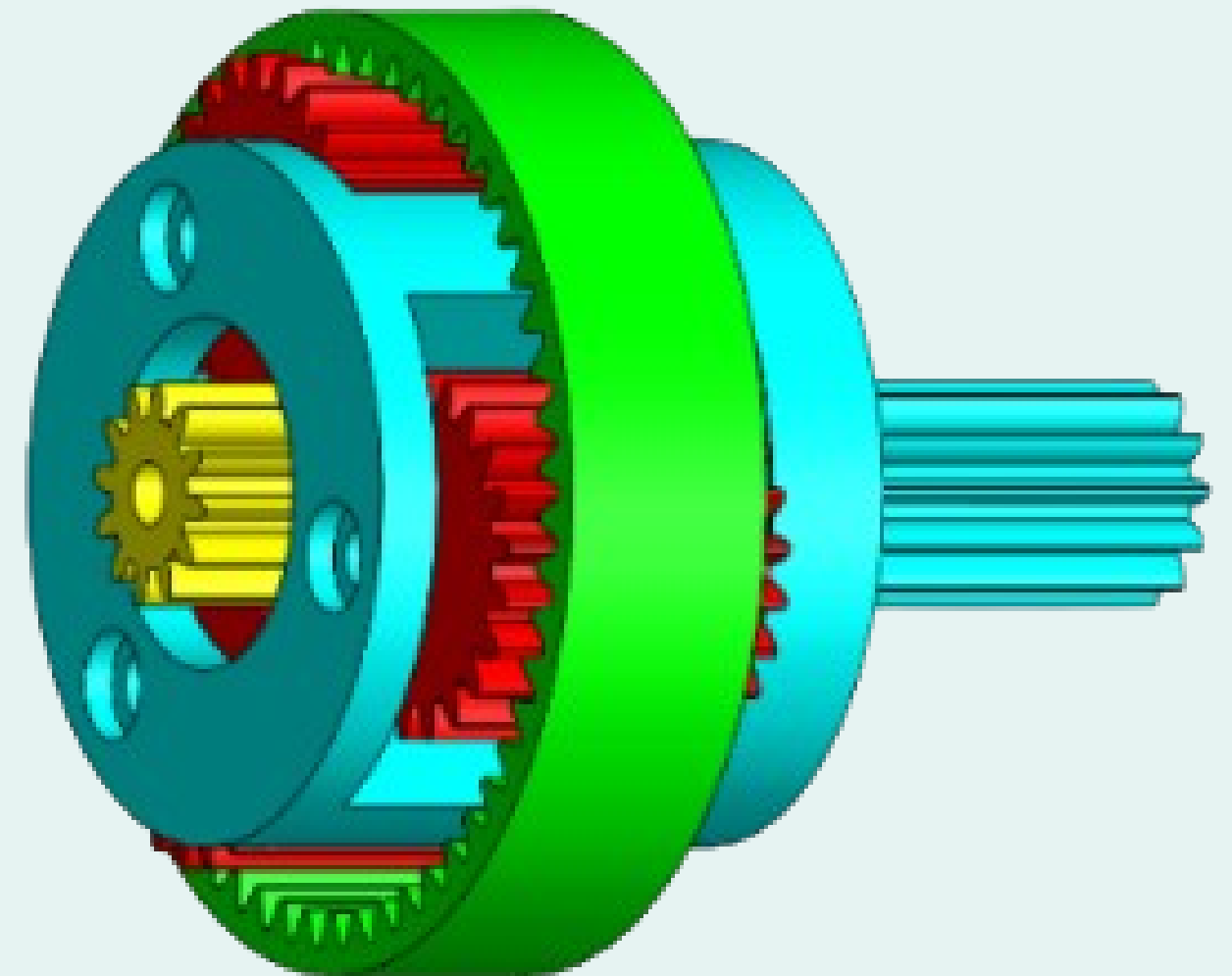
2. PLANETARY GEAR SYSTEM:

$$\text{Gear ratio of one stage of gear system} = 1 + \frac{N_r}{N_s} = 1 + \frac{54}{12} = 5.5$$

here, N_s = no. of teeth on sun gear.

N_p = no. of teeth on ring gear.

$$\text{Gear-Ratio for three stages} = (5.5)^3 : 1 = 166.375 \cdot 1$$



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Working Mechanism and Design

3. LEAD SCREW MECHANISM:

Converts rotational motion into linear displacement with high precision, crucial for controlled bone distraction.

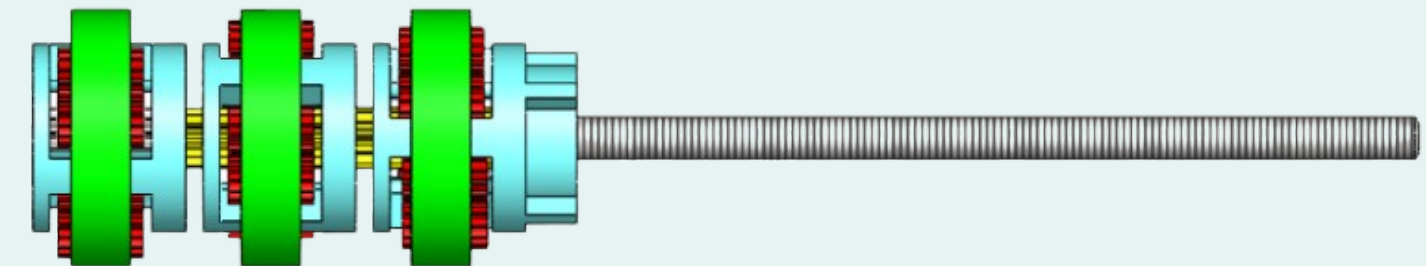
Pitch of lead screw = 1.25 mm per revolution.

Which linearly displace the extuder connected at the lead screw 1.25mm when lead screw rotates one full revolution. It helps to precisely control the lengthening mechanism of the nail as slow as needed.

From the pitch 1.25mm of the lead screw the linear speed of the extruder

$$\begin{aligned} &= (\text{RPM of lead screw}) \times \text{pitch} \\ &= \text{RPM} \times 1.25 \text{ mm per minute} \end{aligned}$$

The desired linear speed of the extruder is 0.3 mm in 2 minutes. From above we can get the required RPM of the motor to get a desired linear speed is around 50 to 80 RPM. Which can be get by PID controlling of the dc motor.



Methodology

PID CONTROL IMPLEMENTATION:

- **Hardware:** Arduino Uno controller, L298N motor driver, Hall-effect encoder.
- **Control algorithm:** Position-based feedback with proportional, integral, and derivative terms.
- **Position accuracy:** High precision through encoder feedback (40,500 pulses per revolution).
- **Direction control** through H-bridge configuration of L298N driver.

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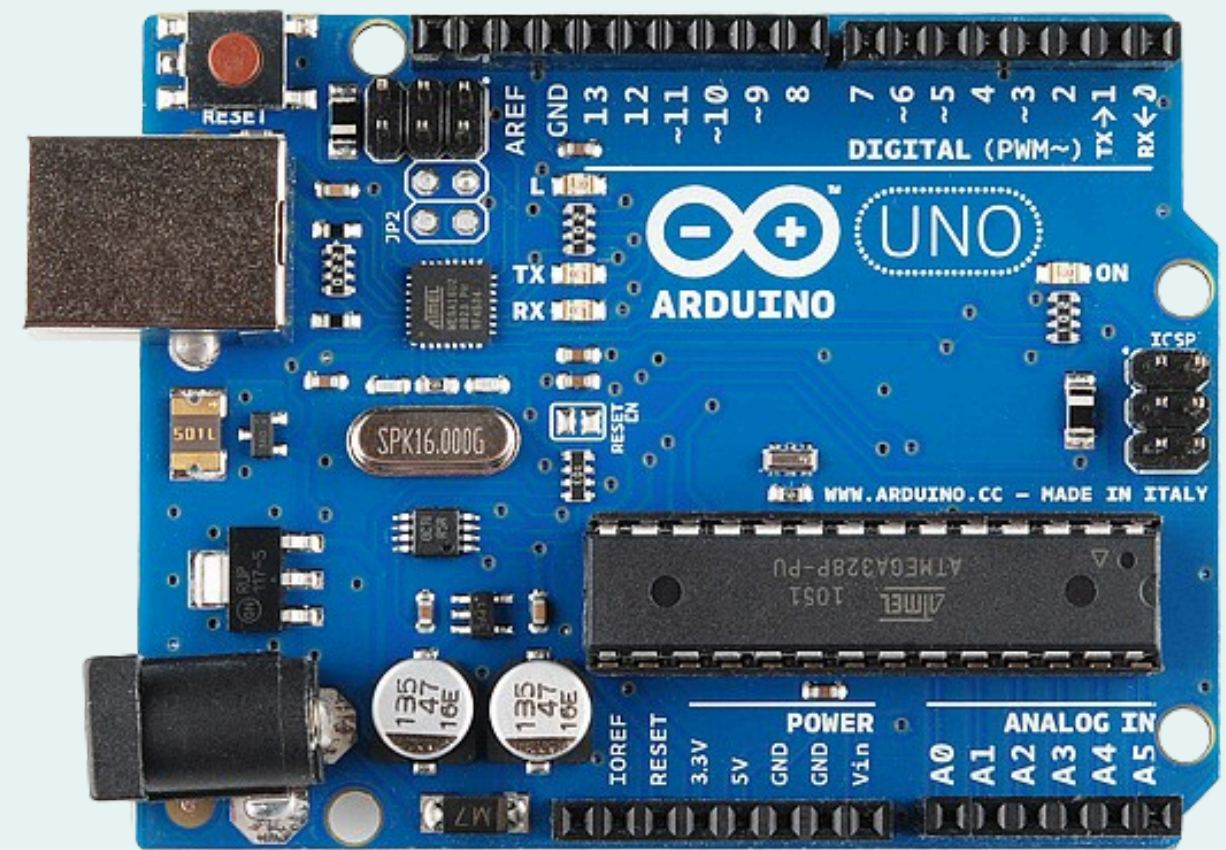
Control System

To accurately control the position of a DC motor, a closed-loop PID control system is implemented. The system uses an Arduino Uno as the controller, an L298N motor driver to control power to the motor, and a Hall-effect encoder for feedback. The PID algorithm continuously adjusts the motor's movement until the desired position is reached with precision.

1. ARDUINO UNO (CONTROLLER):

The Arduino Uno is the brain of the system. It runs a PID control algorithm that continuously calculates the error between the desired position (setpoint) and the current position (from encoder feedback). Based on this error, it computes a control signal to adjust motor power.

Arduino reads the encoder pulses via interrupt pins, calculates position in real time, compares it with the target position, and sends a PWM signal to the L298N accordingly.



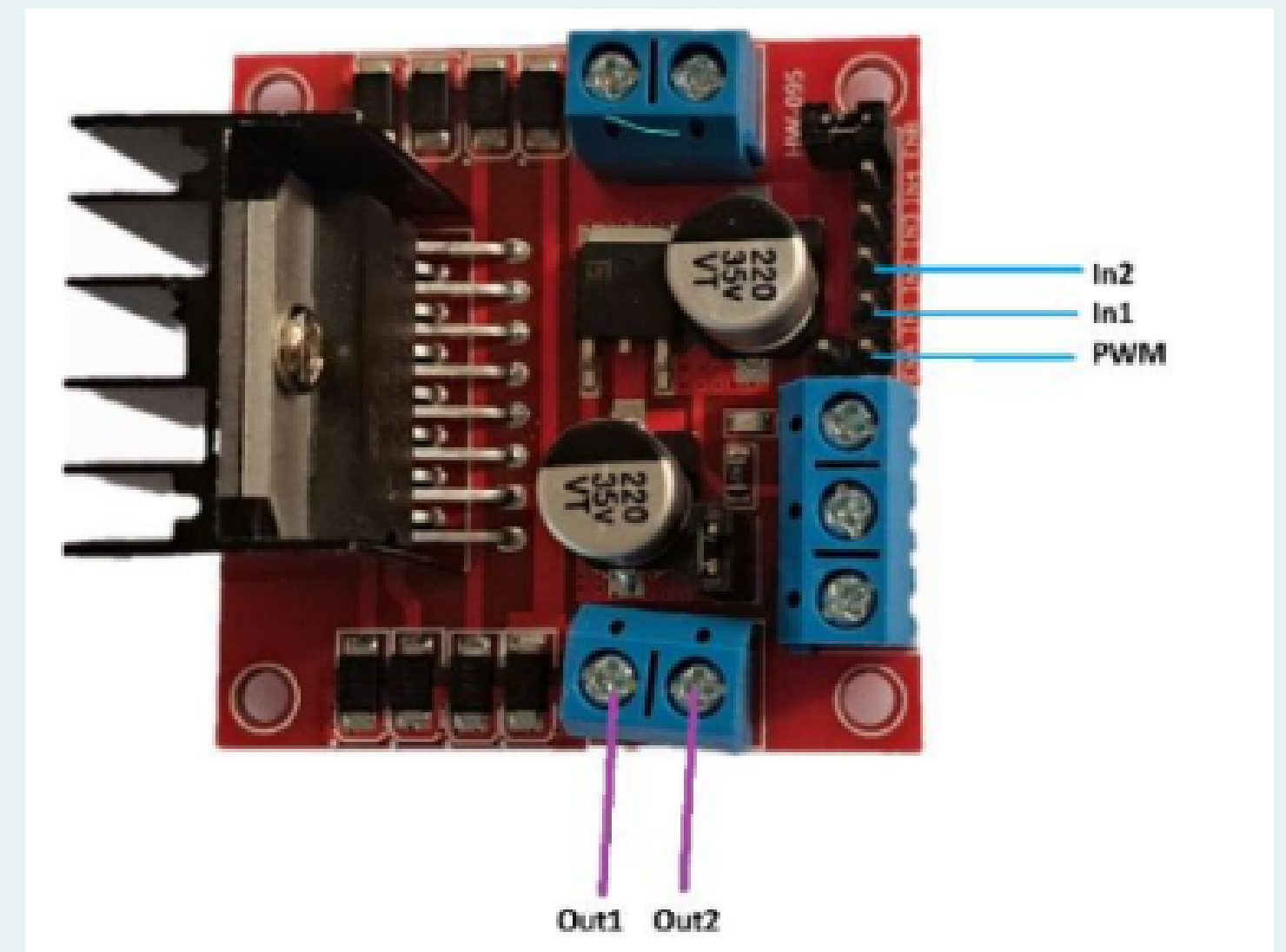
Control System

2.L298N MOTOR DRIVER:

The L298N acts as a bidirectional driver that allows Arduino (which operates at low current) to control the higher-current DC motor.

- **IN1 & IN2**
- **ENA**
- **H-Bridge Structure**

The Arduino sets the direction pins based on desired position (e.g., clockwise for forward, counterclockwise for reverse), and adjusts the PWM duty cycle to control speed based on PID output.

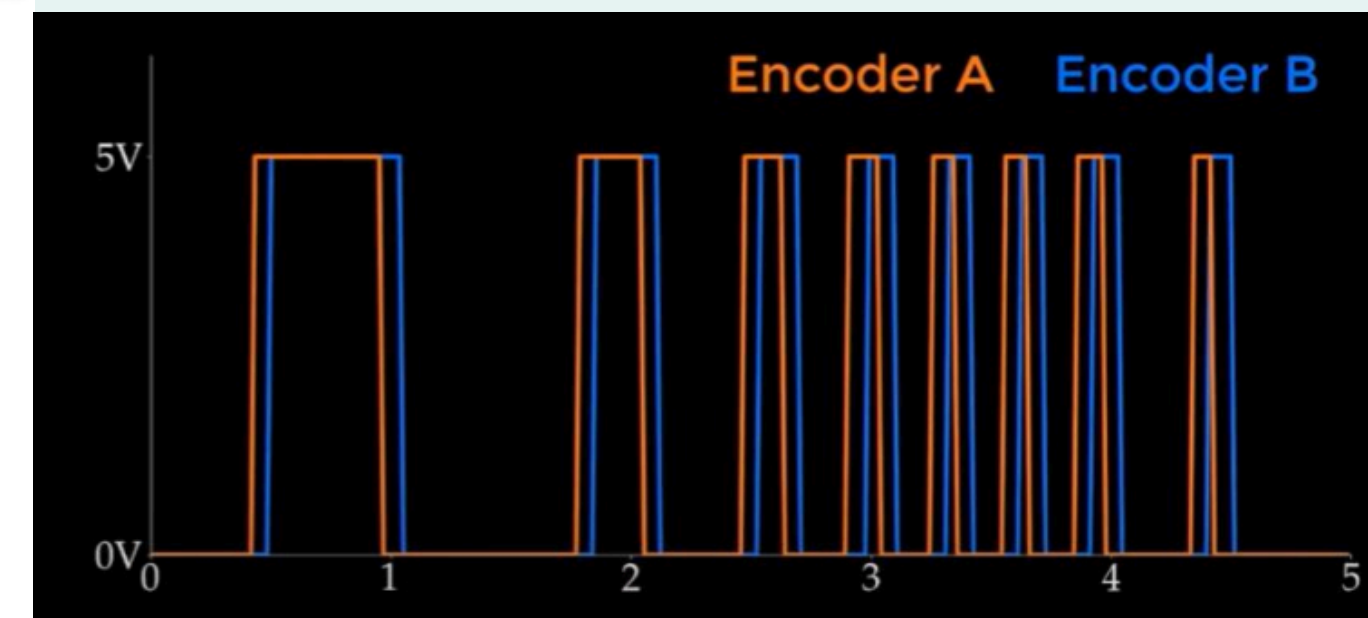
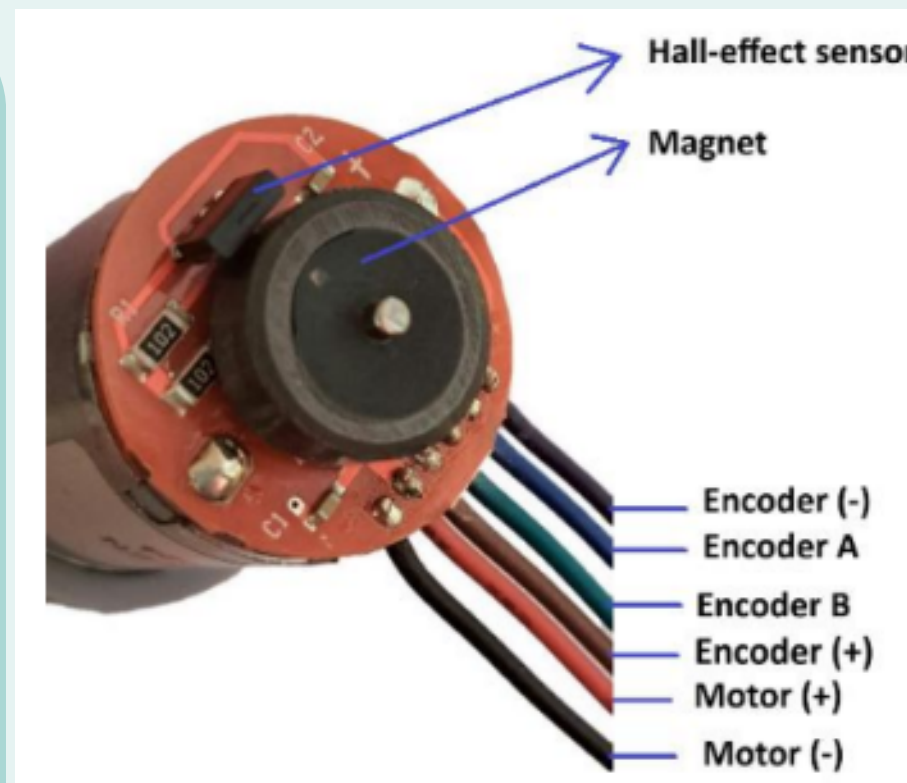


Control System

3. HALL-EFFECT ENCODER (FEEDBACK SENSOR)

The Hall-effect encoder is mounted on the motor shaft and generates a series of digital pulses as the motor rotates. The encoder is consisting of a hall-effect sensor with reads the changes in magnetic field by the magnet attached with shaft.

- i. Encoder (-): Ground signal into the encoder
- ii. Encoder A: Gives the number of pulses sensed by the hall-effect sensor.
- iii. Encoder B: Gives the rotating direction of the motor shaft.
- iv. Encoder (+): 5V into the power the encoder.



Control System

CODES

```
Serial.println("Enter desired angle in degrees:", Do);  
long desiredAngle = Do; // Target angle in degrees  
float currentAngle = 0;
```

Here we give the desired angle of rotation.

$$\text{Output Angle (degrees)} = \left(\frac{\text{Encoder Counts}}{\text{Counts per Motor Rev}} \right) \times \left(\frac{360^\circ}{\text{Gear Ratio}} \right)$$

```
void loop() {  
  // === Calculate current angle ===  
  currentAngle = (encoderCount / COUNTS_PER_REV) * 360.0;  
  
  // === PID Calculations ===  
  error = desiredAngle - currentAngle;  
  integral += error;  
  derivative = error - previousError;  
  
  pidOutput = Kp * error + Ki * integral + Kd * derivative;  
  
  previousError = error;  
  
  // === Motor Control ===  
  setMotor(pidOutput);  
  
  Serial.print("Angle: "); Serial.print(currentAngle);  
  Serial.print(" Error: "); Serial.print(error);  
  Serial.print(" PWM: "); Serial.println(abs(pidOutput));  
  
  delay(10);  
}
```

The feed angle is converted into the no. of pulses from encoder required to reach at a desired position i.e. for one revolution 40500 pulses needed. Then Arduino reads the current number of pulses detected to find the current position. Then a function, error counts the difference between the desired angle and the current angle. Then the PID control function accumulate the error and calculate the output PWM signal for motor driver to set the desired speed of motor and to stop the motor.

$$\text{Control Output} = K_p \cdot \text{Error} + K_i \cdot \text{Integral} + K_d \cdot \text{Derivative}$$

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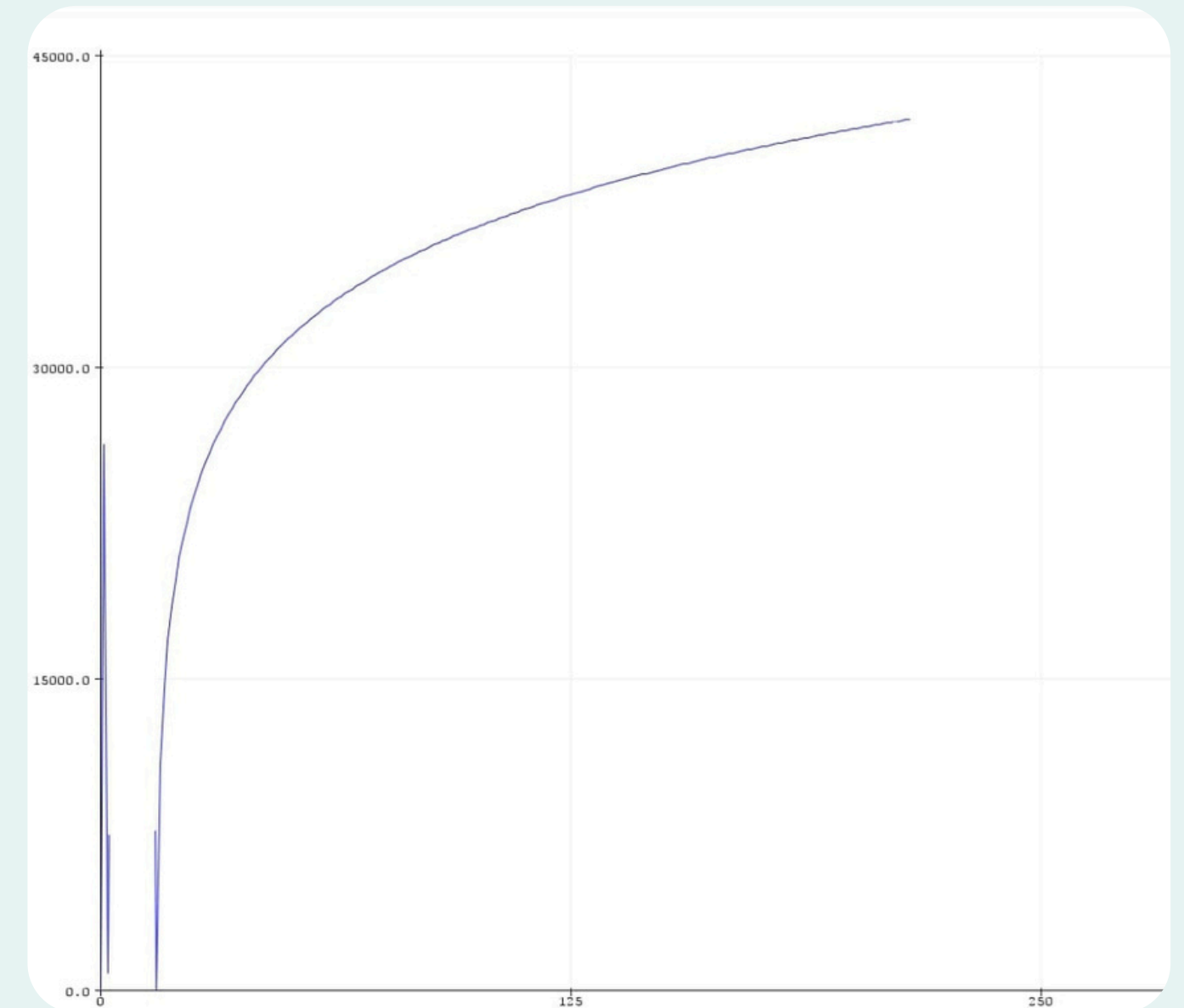
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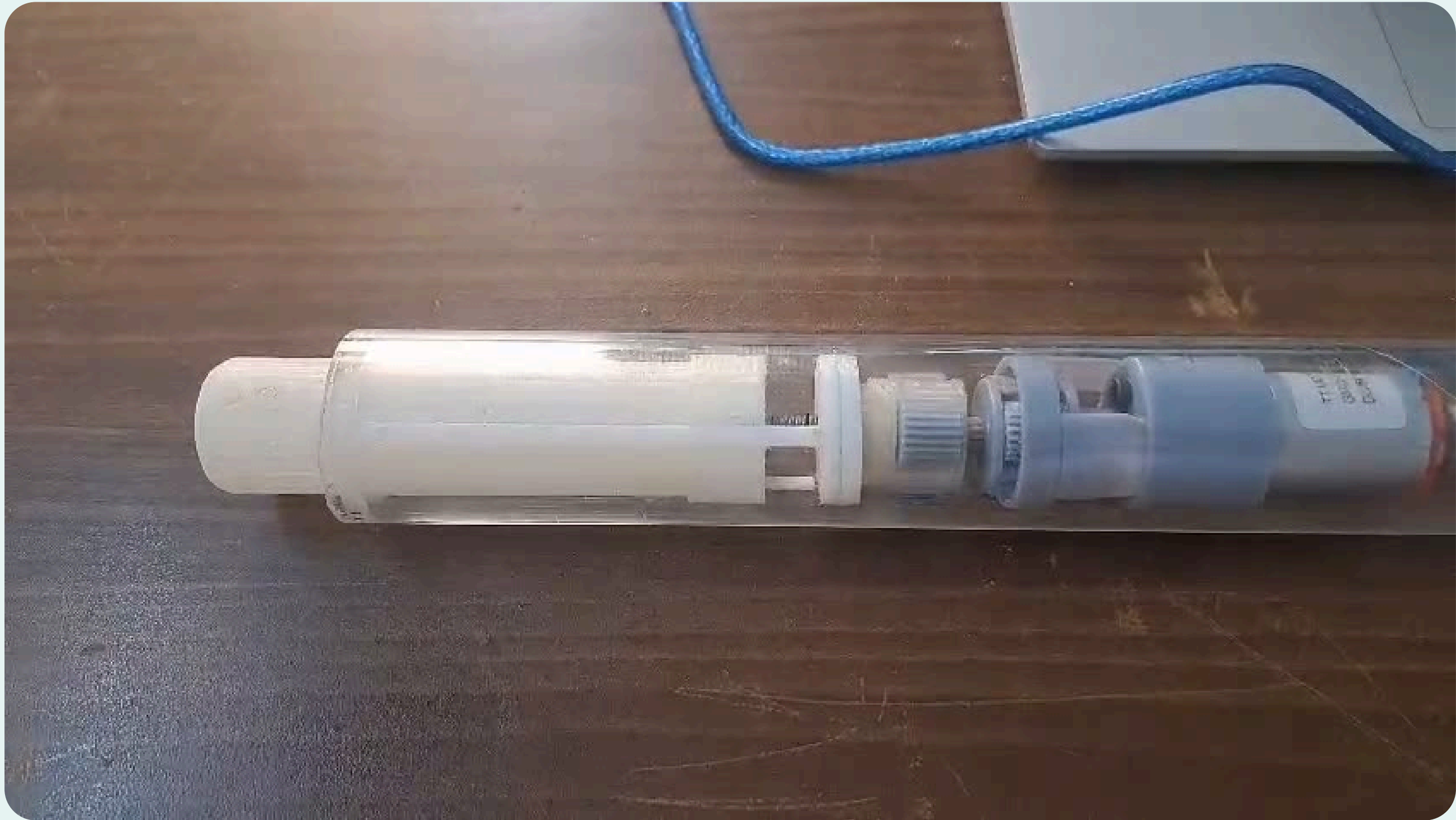
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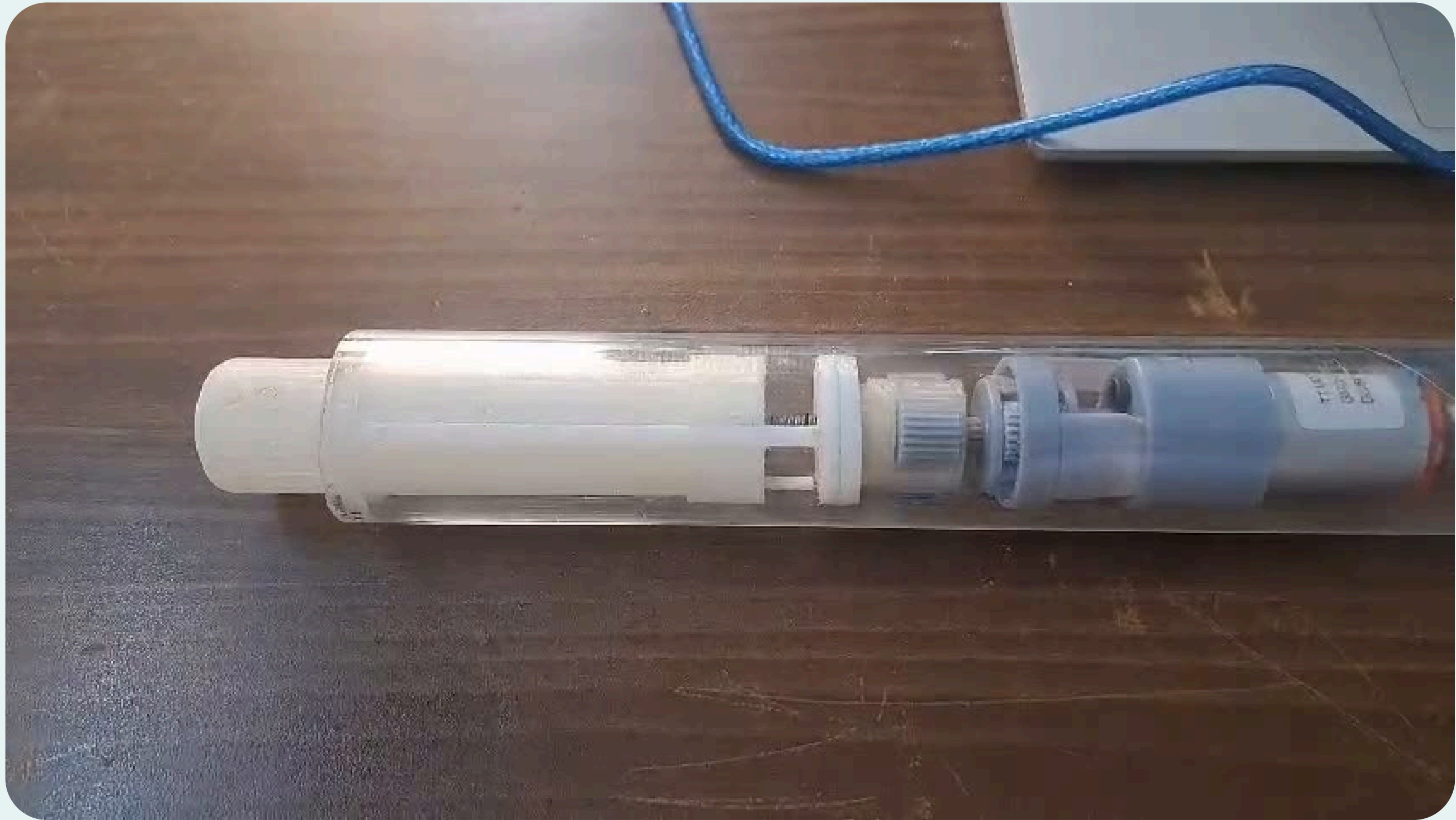
```
void setMotor(float pidValue) {  
    int pwm = constrain(abs(pidValue), 0, 255); // Limit PWM signal  
  
    if (pidValue > 0) {  
        digitalWrite(MOTOR_IN1, HIGH);  
        digitalWrite(MOTOR_IN2, LOW);  
    } else if (pidValue < 0) {  
        digitalWrite(MOTOR_IN1, LOW);  
        digitalWrite(MOTOR_IN2, HIGH);  
    } else {  
        digitalWrite(MOTOR_IN1, LOW);  
        digitalWrite(MOTOR_IN2, LOW);  
    }  
}
```



Final Output



Final Output



**Thank you
very much!**